

The nine-component single-site T-matrix from moment integrals, without Eshelby

Abstract

The cubic single-site T-matrix is usually obtained by importing Eshelby's uniform-eigenstrain result and patching it for a cube. This note derives it instead from the moment integrals of the elastodynamic Green's tensor over the cube. Eshelby is never invoked: it *emerges* as the first-gradient truncation of a closed hierarchy, and the derivation says exactly what that truncation costs. The route is constructive throughout — every integral is in closed form, the $r = 0$ distributional content is carried automatically rather than patched in, and the resulting 9×9 block reduces under O_h to three scalars. Two independent implementations agree on those scalars to 4×10^{-16} .

1 The closed set

Let the scatterer be a cube $V = [-\Delta/2, \Delta/2]^3$ of contrast $\delta c_{ijkl} = \delta\lambda \delta_{ij}\delta_{kl} + \delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and $\delta\rho$, embedded in an isotropic background with Green's tensor G_{ij}^0 . Expanding the internal field about the cube centre and projecting the Lippmann–Schwinger equation onto the monomials $1, x_p, x_px_q$ gives a closed system in the three unknowns $u_i(0), \partial_p u_i(0), \partial_p \partial_q u_i(0)$:

$$(\delta_{ij} - \omega^2 \delta\rho G_{ij}^0) u_j(0) + (N_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \delta\rho K_{ij}^{rs}) \partial_r \partial_s u_j(0) = u_i^0(0), \quad (1)$$

$$(\delta_{pr}\delta_{ij} + \omega^2 \delta\rho N_{ij,p}^r - M_{in,pk} \delta c_{nkrj}) \partial_r u_j(0) = \partial_p u_i^0(0), \quad (2)$$

$$-\omega^2 \delta\rho M_{ij,pq} u_j(0) + (\delta_{qr}\delta_{ps}\delta_{ij} + Q_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \delta\rho P_{ij,pq}^{rs}) \partial_r \partial_s u_j(0) = \partial_p \partial_q u_i^0(0). \quad (3)$$

The six coefficient tensors are moments of G^0 over the cube,

$$\begin{aligned} G_{ij} &= \int_V G_{ij}^0 dV, & K_{ij}^{rs} &= \int_V G_{ij}^0 x_r x_s dV, & N_{in,k}^r &= \int_V \partial_k G_{in}^0 x_r dV, \\ M_{in,pk} &= \int_V \partial_p \partial_k G_{in}^0 dV, & P_{ij,pq}^{rs} &= \int_V \partial_q \partial_p G_{ij}^0 x_r x_s dV, & Q_{in,pqk}^r &= \int_V \partial_q \partial_p \partial_k G_{in}^0 x_r dV, \end{aligned} \quad (4)$$

graded by (D, W) , the number of derivatives and the degree of the x -weight:

$$G(0,0) \quad K(0,2) \quad N(1,1) \quad M(2,0) \quad P(2,2) \quad Q(3,1).$$

A convention that is easy to get backwards. The derivatives in (4) are with respect to the *source* point. Since G^0 depends on the separation, $\partial' = -\partial$ and a moment with D derivatives differs from its field-derivative value by $(-1)^D$. Only N and Q are affected. This is invisible to every symmetry, parity and scaling check, so it must be fixed by convention rather than discovered by testing.

2 The parity decoupling

$D + W$ is even for five of the six moments and odd only for N . Equivalently, $u, \partial u$ and $\partial\partial u$ carry degrees 0, 1, 2, and even couples only to even. The consequence is (2): *it involves nothing else*. The first gradient is determined by its own block, with no reference to u or to $\partial\partial u$.

This is the structural fact the rest of the note rests on. The leading scattering from a modulus contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which needs only ∂u . Therefore *no amount of work on the second-gradient block (3) can change the leading far field*. Whatever (3) contributes reaches the far field only through the $\mathcal{O}((ka)^2)$ coupling in (1).

3 The moments, in closed form

Write the elastodynamic tensor as

$$G_{in}^0(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_\beta(r) + \frac{1}{k_\beta^2} \partial_i \partial_n (g_\beta(r) - g_\alpha(r)) \right], \quad g_k(r) = \frac{e^{ikr}}{r}. \quad (5)$$

Its static limit reproduces Kupradze exactly, with $a_0 = (\lambda + 3\mu)/8\pi\mu(\lambda + 2\mu)$ and $b_0 = (\lambda + \mu)/8\pi\mu(\lambda + 2\mu)$. Expanding g_k in powers of r turns every moment in (4) into a finite sum of *scalar* moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \left\langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \right\rangle. \quad (6)$$

Why these are distributions, not integrals. For $m = -1$ the classical integrand is not absolutely integrable: $\partial^4(1/r)$ is a harmonic, hence traceless, tensor of degree -5 , so its angular average vanishes and the radial divergence is multiplied by zero. The value then depends on the shape of the exclusion. This is precisely the delta-function term that naive quadrature misses — and gets the *sign* wrong without.

The remedy is not to patch a delta in afterwards but to pair against the test function $w \mathbf{1}_V$ from the start. Using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$,

$$\langle \partial_q T', w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T' \, dA - \langle T', (\partial_q w) \mathbf{1}_V \rangle, \quad (7)$$

with $T' = \partial_{\text{rest}} r^m$. Two features make (7) exact rather than merely plausible: T' is evaluated *only on the boundary*, at distance $\Delta/2$ from the origin, where every derivative of r^m is smooth and bounded; and the recursion lowers both D and W , terminating on an ordinary convergent integral. The delta content is inside the surface term and cannot be dropped. The sharp check is

$$\sum_p E[-1; \{p, p\};] = -4\pi, \quad (8)$$

which a spherical excision returns as 0.

Conditioning. The integrals emit logarithms of Pell units, $(2 \pm \sqrt{3})^n = A_n \pm B_n \sqrt{3}$ solving $x^2 - 3y^2 = 1$. Left unreduced these are numerically lethal: $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a 35-digit cancellation between integers of order 10^{17} and evaluates to **Indeterminate** in double precision. Reduced by $\log(A_n - B_n \sqrt{3}) = n \log(2 - \sqrt{3})$ it is exact to machine precision. Every closed form below is Pell-reduced.

4 The nine-component block

Specialise (2). With $\delta\rho = 0$ — the regime in which the single-site modulus T-matrix is defined — the block is

$$A_{(p,i),(r,j)}^{(2)} = \delta_{pr} \delta_{ij} - M_{in,pk} \delta_{nkrj}, \quad (9)$$

a 9×9 operator built from M alone. It contains neither Q nor P : the second-gradient moments do not appear.

Under O_h the nine components decompose as $\mathbf{3} \otimes \mathbf{3} = A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$, and (9) is diagonal in that basis. The strain concentration in each channel is the inverse of the corresponding eigenvalue:

$$A_{1g} \text{ (bulk)} : 1 + \frac{3\delta\lambda + 2\delta\mu}{3(\lambda + 2\mu)}, \quad (10)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\delta\mu S_{\text{shear}}, \quad S_{\text{shear}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (11)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\mu\pi}{6\pi\mu(\lambda + 2\mu)}. \quad (12)$$

That $E_g \neq T_{2g}$ is not a discrepancy: they are inequivalent representations, and their difference *is* the cubic anisotropy, carried elsewhere in this project as $\Delta\mu_{\text{diag}}^* - \Delta\mu_{\text{off}}^*$. A sphere would have no such split; a cube must.

5 Where Eshelby went

Eshelby’s uniformity theorem says the internal field of an inclusion under uniform remote strain is uniform *if and only if* the inclusion is an ellipsoid. The usual construction imports the ellipsoidal result and adapts it. Nothing of the sort happens here. Equations (10)–(12) follow from the moment integrals and the group theory alone.

What *does* correspond to Eshelby is the truncation: retaining only ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz. So the classical result is recovered as the first-gradient closure of (1)–(3), and the hierarchy makes the cost of that closure explicit rather than hidden.

And the cost does not shrink by going further. Because (2) is self-contained, adding the second-gradient block cannot improve the leading far field. This is not an estimate; it is a consequence of the parity decoupling. It also explains an otherwise puzzling observation: the tier sequence $T_9 \rightarrow T_{27} \rightarrow T_{57}$ in the shear channel runs $1.000 \rightarrow 0.7525 \rightarrow 0.9692$ — oscillating, not converging. A polynomial expansion about the centre cannot represent the edge and corner stress singularities of a cube at any order, so the series is asymptotic. Improving the far field requires changing the *basis*, not adding terms to it.

6 Validation

The construction is checked against objects it does not contain.

Check	Against	Result
$\sum_p E[-1; \{p, p\}] = -4\pi$	analytic	exact
$\nabla^2 r = 2/r$, $\nabla^2 r^3 = 12r$, $\nabla^2 r^5 = 30r^3$	analytic	exact
$\int_V dV/r$	solid-angle quadrature	3.4×10^{-9}
M components A, B, C	independent first-principles route	10 digits
M , all 81 components	2012 archive closed form	exact
T_2^c, T_3^c	production <code>effective_contrasts</code>	3.6×10^{-16} , 1.3×10^{-15}

The last row is the load-bearing one. The production code reaches the same quantities by a completely different route — tabulated master integrals with an explicit Eshelby delta correction

— and the two agree to machine precision. Matching the identification

$$\text{amp}_e^{\text{off}} = \frac{1}{1 - 2T_2}, \quad \text{amp}_e^{\text{diag}} = \frac{1}{1 - 2T_2 - T_3}$$

against (11)–(12) predicts, with no free parameters,

$$T_2^c = -\delta\mu S_{\text{shear}}, \quad T_3^c = 2\delta\mu (S_{\text{shear}} - S_{\text{diag}}),$$

and both hold to 10^{-15} . Two independent implementations agreeing is the evidence standard; one implementation agreeing with itself is not.

7 What this does and does not establish

Established. The nine-component single-site T-matrix follows from the moment integrals with no ellipsoidal input; the moments are exact and carry their distributional content correctly; the block reduces to three closed-form scalars; and those scalars are confirmed by an independent implementation.

Not established. That the result is *accurate* for a cube. It is the first-gradient truncation of an asymptotic hierarchy, and (10)–(12) inherit whatever error that truncation carries. The construction makes the approximation explicit and bounds where it can live — in the internal-field basis, not in the integrals — but it does not remove it. Improving on it means replacing the polynomial ansatz, for instance by supplying the three channel concentrations from an independent exact computation while keeping everything else in this note intact.